

PROBLEM SET 5

Show **ALL WORK** to get full/partial credit. Begin each problem on a new page, and clearly label each part of the problem. (**Max Score:** 90)

1) [Shankar 17.2.3] **(30 pts)** In our study of the H atom, we assumed that the proton is a point charge e . This leads to the familiar Coulomb interaction ($V^0 = -e^2/r$) with the electron, which we can treat as a *zeroth*-order perturbation.

(a) **(10 pts)** Show that if the proton is a *uniformly* dense charge distribution of radius $R \ll a_0$, the electric field is

$$E(r) = \begin{cases} er/R^3 & (r < R) \\ e/r^2 & (r \geq R) \end{cases}$$

hint: use Gauss's law $\oint_S \mathbf{E} \cdot d\mathbf{A} = Q_{\text{encl}}$ ($\epsilon_0 = 1$ is cgs units)

(b) **(10 pts)** Show that the corresponding electric potential energy is

$$V(r) = -e\phi(r) = \begin{cases} -\frac{3e^2}{2R} + \frac{e^2 r^2}{2R^3} & (r < R) \\ -\frac{e^2}{r} & (r \geq R) \end{cases}$$

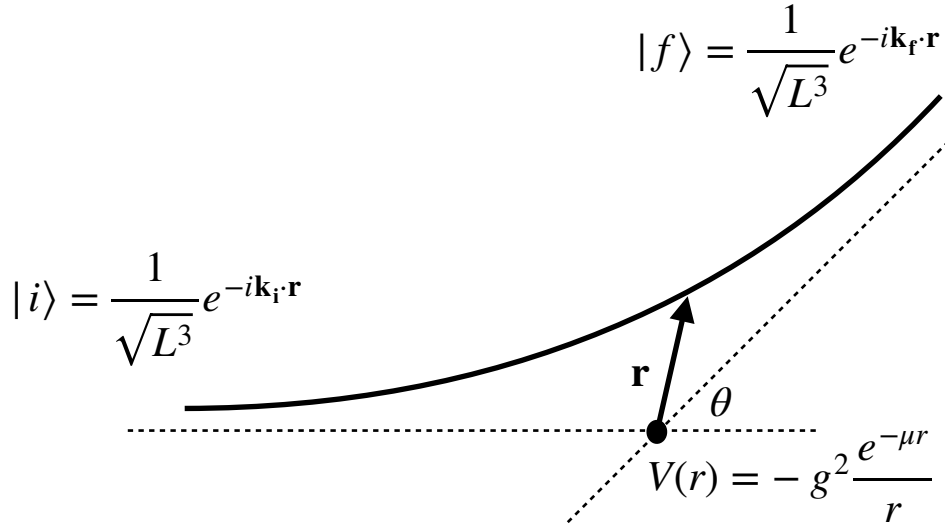
hint: $\mathbf{E} = -\nabla\phi$

(c) **(10 pts)** Using the 1st order perturbation theory, $E^1 = \langle n^0 | H^1 | n^0 \rangle$, and the 1st order perturbation ($r < R$) determined in part (b), to calculate the first-order shift in the ground-state energy of hydrogen due to this modification and show it is $E^1 \approx \frac{2}{5} \frac{e^2 R^2}{a_0^3}$

hint: ground-state hydrogen is given by $|n^0\rangle \equiv |100\rangle = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0}$

Since $R \ll a_0$, you may assume $e^{-r/a_0} \sim 1$

- 2) **(30 pts)** A free particle (normalized to the volume L^3 of a box) scatters off at an angle θ from a Yukawa potential, $V(r) = -g^2 \frac{e^{-\mu r}}{r}$, where $\mu = m_\pi c / \hbar$ with m_π representing the mass of the exchanged particle between the nucleons, responsible for the long-range part of the nuclear force, and $-g^2$ is a strong coupling constant where the “-” sign reflects the attractive nature of the nuclear force.



- (a) **(5 pts)** Show that the transition matrix V_{fi} in *Fermi's golden rule* is a Fourier transform of the potential from position to momentum space which has the form

$$V_{fi} \equiv V(q) = \frac{1}{L^3} \int V(r) e^{-i\mathbf{q} \cdot \mathbf{r} / \hbar} d^3\mathbf{r},$$

where $\mathbf{q} \equiv \mathbf{p}_i - \mathbf{p}_f = \hbar(\mathbf{k}_i - \mathbf{k}_f)$ is the momentum transfer

(b) **(10 pts)** Compute the integral from part (a) to show the transition matrix element has the general form $V(q) = -\frac{4\pi\hbar^2}{L^3} \frac{g^2}{|\mathbf{q}|^2 + m_\pi^2 c^2}$

hint: assume $\mathbf{q}$ is along the $+z$ axis to simplify the integral in spherical coordinates, such that $\mathbf{q} \cdot \mathbf{r} = qr \cos \theta$

useful integrals: $\int_0^\pi e^{-iqr \cos \theta / \hbar} \sin \theta d\theta = \frac{2\hbar}{qr} \sin(qr/\hbar),$

$$\int_0^\infty e^{-ar} \sin(br) dr = \frac{b}{a^2 + b^2}$$

(c) **(5 pts)** For the transition into a continuum of final states subtended by a solid angle $d\Omega$ into a detector, Fermi's golden rule is states as follows: $R_{i \rightarrow f} = \frac{2\pi}{\hbar} |\langle f | V(r) | i \rangle|^2 \rho(E_f)$, where the density of states is given by $\rho(E_f) = \frac{L^3}{(2\pi)^3} \frac{mk_f}{\hbar^2} d\Omega$ with incident particle mass m .

Show that the detection rate (or scattered flux) into solid angle is given by

$$\frac{R_{i \rightarrow d\Omega}}{d\Omega} = \frac{4\hbar g^4 m k_f}{L^3 (|\mathbf{q}|^2 + m_\pi^2 c^2)^2}$$

- (d) **(10 pts)** The differential cross section is defined as the ratio of the scattered to incident current densities (or flux), defined as

$$\frac{d\sigma}{d\Omega} \equiv \frac{R_{i \rightarrow d\Omega} / d\Omega}{j_{\text{inc}}}, \text{ where the incident particle (of mass } m) \text{ flux is given}$$

$$\text{by } j_{\text{inc}} = \frac{\hbar k_i}{mL^3}$$

Show that the final differential cross section can be expressed as

$$\frac{d\sigma}{d\Omega} = \frac{4m^2 g^4}{[4\hbar^2 k^2 \sin^2(\theta/2) + m_\pi^2 c^2]^2}$$

hint: (1) assume elastic scattering and use the fact that the incident and scattered free particle energies are conserved which implies $k_i = k_f \equiv k$, (2) use the fact the angle between the incident ($\hbar \mathbf{k}_i$) and scattered ($\hbar \mathbf{k}_f$) momenta is θ to express $|\mathbf{q}|^2 = |\hbar \mathbf{k}_i - \hbar \mathbf{k}_f|^2$ in terms of the angle and momenta.

3) [Shankar 19.3.3] **(30 pts)** For spherically-symmetric potentials, the general expression for the scattering amplitude is given by

$$f(\theta) = \frac{-2\mu}{\hbar^2} \int \frac{\sin qr'}{q} V(r') r' dr'$$

Show that for the Gaussian potential, $V(r) = V_0 e^{-r^2/r_0^2}$:

(a) **(15 pts)** The differential cross section is given by:

$$\frac{d\sigma}{d\Omega} = \frac{\pi r_0^2}{4} \left(\frac{\mu V_0 r_0^2}{\hbar^2} \right)^2 e^{-q^2 r_0^2 / 2}$$

useful integral: $\int_0^\infty \sin(ax) e^{-x^2/b^2} x dx = \frac{\sqrt{\pi} a b^3}{4} e^{-a^2 b^2 / 4}$

(b) **(15 pts)** The total cross section is given by:

$$\sigma = \frac{\pi^2}{2k^2} \left(\frac{\mu V_0 r_0^2}{\hbar^2} \right)^2 (1 - e^{-2k^2 r_0^2})$$

hint: Since $q^2 = 2k^2(1 - \cos \theta)$, $d(\cos \theta) = -d(q^2)/2k^2$,

useful integral: $\int_{-1}^1 e^{ax} dx = \frac{e^a - e^{-a}}{a}$